# CHAPTER 3 SUMMARY

**Position, velocity, and acceleration vectors:** The position vector $\vec{r}$ of a point $P$ in space is the vector from the origin to $P$. Its components are the coordinates $x$, $y$, and $z$.

The average velocity vector $\vec{v}_{av}$ during the time interval $\Delta t$ is the displacement $\Delta\vec{r}$ (the change in position vector $\vec{r}$) divided by $\Delta t$. The instantaneous velocity vector $\vec{v}$ is the time derivative of $\vec{r}$, and its components are the time derivatives of $x$, $y$, and $z$. The instantaneous speed is the magnitude of $\vec{v}$. The velocity $\vec{v}$ of a particle is always tangent to the particle's path. (See Example 3.1.)

The average acceleration vector $\vec{a}_{av}$ during the time interval $\Delta t$ equals $\Delta\vec{v}$ (the change in velocity vector $\vec{v}$) divided by $\Delta t$. The instantaneous acceleration vector $\vec{a}$ is the time derivative of $\vec{v}$, and its components are the time derivatives of $v_x$, $v_y$, and $v_z$. (See Example 3.2.)

The component of acceleration parallel to the direction of the instantaneous velocity affects the speed, while the component of $\vec{a}$ perpendicular to $\vec{v}$ affects the direction of motion. (See Examples 3.3 and 3.4.)

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k} \quad (3.1)$$

$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\Delta\vec{r}}{\Delta t} \quad (3.2)$$

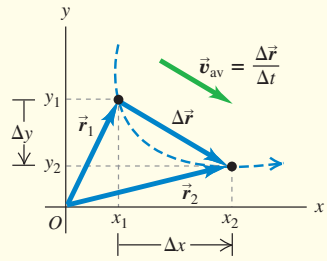
$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} \quad (3.3)$$

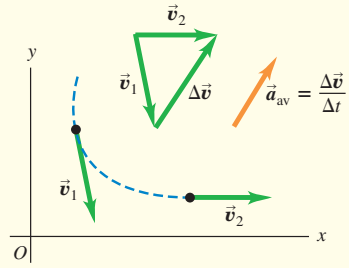
$$v_x = \frac{dx}{dt} \quad v_y = \frac{dy}{dt} \quad v_z = \frac{dz}{dt} \quad (3.4)$$

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\Delta\vec{v}}{\Delta t} \quad (3.8)$$

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} \quad (3.9)$$

$$a_x = \frac{dv_x}{dt}, \quad a_y = \frac{dv_y}{dt}, \quad a_z = \frac{dv_z}{dt} \quad (3.10)$$



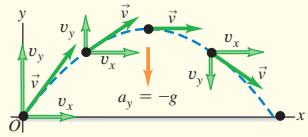

**Projectile motion:** In projectile motion with no air resistance, $a_x = 0$ and $a_y = -g$. The coordinates and velocity components are simple functions of time, and the shape of the path is always a parabola. We usually choose the origin to be at the initial position of the projectile. (See Examples 3.5 –3.10.)

$$x = (v_0 \cos\alpha_0)t \quad (3.19)$$

$$y = (v_0 \sin\alpha_0)t - \tfrac{1}{2}gt^2 \quad (3.20)$$

$$v_x = v_0 \cos\alpha_0 \quad (3.21)$$

$$v_y = v_0 \sin\alpha_0 - gt \quad (3.22)$$

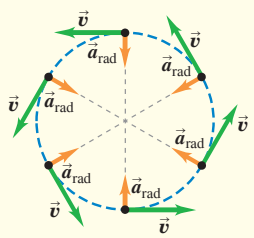



**Uniform and nonuniform circular motion:** When a particle moves in a circular path of radius $R$ with constant speed $v$ (uniform circular motion), its acceleration $\vec{a}$ is directed toward the center of the circle and perpendicular to $\vec{v}$. The magnitude $a_{rad}$ of this radial acceleration can be expressed in terms of $v$ and $R$ or in terms of $R$ and the period $T$ (the time for one revolution), where $v = 2\pi R/T$. (See Examples 3.11 and 3.12.)

If the speed is not constant in circular motion (nonuniform circular motion), there is still a radial component of $\vec{a}$ given by Eq. (3.27) or (3.29), but there is also a component of $\vec{a}$ parallel (tangential) to the path. This tangential component is equal to the rate of change of speed, $dv/dt$.

$$a_{rad} = \frac{v^2}{R} \quad (3.27)$$

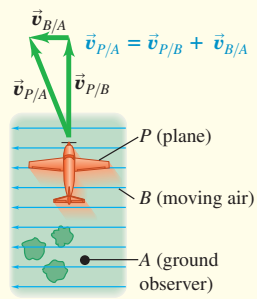
$$a_{rad} = \frac{4\pi^2 R}{T^2} \quad (3.29)$$




**Relative velocity:** When an object $P$ moves relative to an object (or reference frame) $B$, and $B$ moves relative to an object (or reference frame) $A$, we denote the velocity of $P$ relative to $B$ by $\vec{v}_{P/B}$, the velocity of $P$ relative to $A$ by $\vec{v}_{P/A}$, and the velocity of $B$ relative to $A$ by $\vec{v}_{B/A}$. If these velocities are all along the same line, their components along that line are related by Eq. (3.32). More generally, these velocities are related by Eq. (3.35). (See Examples 3.13 –3.15.)

$$v_{P/A\text{-}x} = v_{P/B\text{-}x} + v_{B/A\text{-}x} \quad \text{(relative velocity along a line)} \quad (3.32)$$

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A} \quad \text{(relative velocity in space)} \quad (3.35)$$



Chapter 3 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

## GUIDED PRACTICE

### KEY EXAMPLE VARIATION PROBLEMS

**Be sure to review EXAMPLES 3.6, 3.7, 3.8, and 3.9 (Section 3.3) before attempting these problems. In all problems, ignore air resistance.**

**VP3.9.1** You launch a projectile from level ground at a speed of 25.0 m/s and an angle of 36.9° above the horizontal. (a) How long after it is launched does the projectile reach its maximum height above the ground, and what is that maximum height? (b) How long after the projectile is launched does it return to ground level, and how far from its launch point does it land?

**VP3.9.2** You throw a baseball at an angle of 30.0° above the horizontal. It reaches the highest point of its trajectory 1.05 s later. (a) At what speed does the baseball leave your hand? (b) What is the maximum height above the launch point that the baseball reaches?

**VP3.9.3** You toss a walnut at a speed of 15.0 m/s at an angle of 50.0° above the horizontal. The launch point is on the roof of a building that is 20.0 m above the ground. (a) How long after it is launched does the walnut reach the ground? (b) How far does the walnut travel horizontally from launch point to landing point? (c) What are the horizontal and vertical components of the walnut's velocity just before it reaches the ground?

**VP3.9.4** You use a slingshot to launch a potato horizontally from the edge of a cliff with speed $v_0$. The acceleration due to gravity is $g$. Take the origin at the launch point. (a) How long after you launch the potato has it moved as far horizontally from the launch point as it has moved vertically? What are the coordinates of the potato at this time? (b) How long after you launch the potato is it moving in a direction exactly 45° below the horizontal? What are the coordinates of the potato at this time?

**Be sure to review EXAMPLES 3.11 and 3.12 (Section 3.4) before attempting these problems.**

**VP3.12.1** A cyclist going around a circular track at 10.0 m/s has a centripetal acceleration of 5.00 m/s$^2$. What is the radius of the curve?

**VP3.12.2** A race car is moving at 40.0 m/s around a circular racetrack of radius 265 m. Calculate (a) the period of the motion and (b) the car's centripetal acceleration.

**VP3.12.3** The wheel of a stationary exercise bicycle at your gym makes one rotation in 0.670 s. Consider two points on this wheel: Point $P$ is 10.0 cm from the rotation axis, and point $Q$ is 20.0 cm from the rotation axis. Find (a) the speed of each point on the spinning wheel and (b) the centripetal acceleration of each point. (c) For points on this spinning wheel, as the distance from the axis increases, does the speed increase or decrease? Does the centripetal acceleration increase or decrease?

**VP3.12.4** The planets Venus, Earth, and Mars all move in approximately circular orbits around the sun. Use the data in the table to find (a) the speed of each planet in its orbit and (b) the centripetal acceleration of each planet. (c) As the size of a planet's orbit increases, does the speed increase or decrease? Does the centripetal acceleration increase or decrease?

| Planet | Orbital radius (m) | Orbital period (days) |
|---|---|---|
| Venus | $1.08 \times 10^{11}$ | 225 |
| Earth | $1.50 \times 10^{11}$ | 365 |
| Mars | $2.28 \times 10^{11}$ | 687 |

**VP3.12.5** Object $A$ is moving at speed $v$ in a circle of radius $R$. Object $B$ is moving at speed $2v$ in a circle of radius $R/2$. (a) What is the ratio of the period of object $A$ to the period of object $B$? (b) What is the ratio of the centripetal acceleration of object $A$ to the centripetal acceleration of object $B$?

**Be sure to review EXAMPLES 3.13, 3.14, and 3.15 (Section 3.5) before attempting these problems.**

**VP3.15.1** A police car in a high-speed chase is traveling north on a two-lane highway at 35.0 m/s. In the southbound lane of the same highway, an SUV is moving at 18.0 m/s. Take the positive $x$-direction to be toward the north. Find the $x$-velocity of (a) the police car relative to the SUV and (b) the SUV relative to the police car.

**VP3.15.2** Race cars $A$ and $B$ are driving on the same circular racetrack at the same speed of 45.0 m/s. At a given instant car $A$ is on the north side of the track moving eastward and car $B$ is on the south side of the track moving westward. Find the velocity vector (magnitude and direction) of (a) car $A$ relative to car $B$ and (b) car $B$ relative to car $A$. (c) Does the relative velocity have a component along the line connecting the two cars? Are the two cars approaching each other, moving away from each other, or neither?

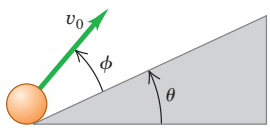
**VP3.15.3** Two vehicles approach an intersection: a truck moving eastbound at 16.0 m/s and an SUV moving southbound at 20.0 m/s. Find the velocity vector (magnitude and direction) of (a) the truck relative to the SUV and (b) the SUV relative to the truck.

**VP3.15.4** A jet is flying due north relative to the ground. The speed of the jet relative to the ground is 155 m/s. The wind at the jet's altitude is 40.0 m/s toward the northeast (45.0° north of east). Find the speed of the jet relative to the air (its airspeed) and the direction in which the pilot of the jet must point the plane so that it travels due north relative to the ground.

### BRIDGING PROBLEM Launching Up an Incline

You fire a ball with an initial speed $v_0$ at an angle $\phi$ above the surface of an incline, which is itself inclined at an angle $\theta$ above the horizontal (**Fig. 3.37**). (a) Find the distance, measured along the incline, from the launch point to the point when the ball strikes the incline. (b) What angle $\phi$ gives the maximum range, measured along the incline? Ignore air resistance.

Figure 3.37 Launching a ball from an inclined ramp.




**SOLUTION GUIDE**

**IDENTIFY and SET UP**

1. Since there's no air resistance, this is a problem in projectile motion. The goal is to find the point where the ball's parabolic trajectory intersects the incline.

*Continued*

2. Choose the $x$- and $y$-axes and the position of the origin. When in doubt, use the suggestions given in Problem-Solving Strategy 3.1 in Section 3.3.
3. In the projectile equations in Section 3.3, the launch angle $\alpha_0$ is measured from the horizontal. What is this angle in terms of $\theta$ and $\phi$? What are the initial $x$- and $y$-components of the ball's initial velocity?
4. You'll need to write an equation that relates $x$ and $y$ for points along the incline. What is this equation? (This takes just geometry and trigonometry, not physics.)

**EXECUTE**

5. Write the equations for the $x$-coordinate and $y$-coordinate of the ball as functions of time $t$.
6. When the ball hits the incline, $x$ and $y$ are related by the equation that you found in step 4. Based on this, at what time $t$ does the ball hit the incline?
7. Based on your answer from step 6, at what coordinates $x$ and $y$ does the ball land on the incline? How far is this point from the launch point?
8. What value of $\phi$ gives the *maximum* distance from the launch point to the landing point? (Use your knowledge of calculus.)

**EVALUATE**

9. Check your answers for the case $\theta = 0$, which corresponds to the incline being horizontal rather than tilted. (You already know the answers for this case. Do you know why?)

## PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

### DISCUSSION QUESTIONS

**Q3.1** A simple pendulum (a mass swinging at the end of a string) swings back and forth in a circular arc. What is the direction of the acceleration of the mass when it is at the ends of the swing? At the midpoint? In each case, explain how you obtained your answer.

**Q3.2** Redraw Fig. 3.11a if $\vec{a}$ is antiparallel to $\vec{v}_1$. Does the particle move in a straight line? What happens to its speed?

**Q3.3** A projectile moves in a parabolic path without air resistance. Is there any point at which $\vec{a}$ is parallel to $\vec{v}$? Perpendicular to $\vec{v}$? Explain.

**Q3.4** A book slides off a horizontal tabletop. As it leaves the table's edge, the book has a horizontal velocity of magnitude $v_0$. The book strikes the floor in time $t$. If the initial velocity of the book is doubled to $2v_0$, what happens to (a) the time the book is in the air, (b) the horizontal distance the book travels while it is in the air, and (c) the speed of the book just before it reaches the floor? In particular, does each of these quantities stay the same, double, or change in another way? Explain.

**Q3.5** At the instant that you fire a bullet horizontally from a rifle, you drop a bullet from the height of the gun barrel. If there is no air resistance, which bullet hits the level ground first? Explain.

**Q3.6** A package falls out of an airplane that is flying in a straight line at a constant altitude and speed. If you ignore air resistance, what would be the path of the package as observed by the pilot? As observed by a person on the ground?

**Q3.7** Sketch the six graphs of the $x$- and $y$-components of position, velocity, and acceleration versus time for projectile motion with $x_0 = y_0 = 0$ and $0 < \alpha_0 < 90°$.

**Q3.8** If a jumping frog can give itself the same initial speed regardless of the direction in which it jumps (forward or straight up), how is the maximum vertical height to which it can jump related to its maximum horizontal range $R_{max} = v_0^2/g$?

**Q3.9** A projectile is fired upward at an angle $\theta$ above the horizontal with an initial speed $v_0$. At its maximum height, what are its velocity vector, its speed, and its acceleration vector?

**Q3.10** In uniform circular motion, what are the *average* velocity and *average* acceleration for one revolution? Explain.

**Q3.11** In uniform circular motion, how does the acceleration change when the speed is increased by a factor of 3? When the radius is decreased by a factor of 2?

**Q3.12** In uniform circular motion, the acceleration is perpendicular to the velocity at every instant. Is this true when the motion is not uniform—that is, when the speed is not constant?

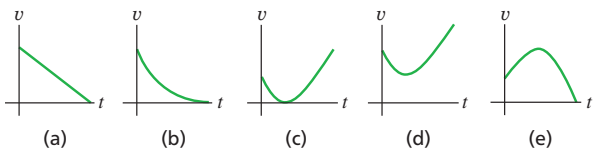
**Q3.13** Raindrops hitting the side windows of a car in motion often leave diagonal streaks even if there is no wind. Why? Is the explanation the same or different for diagonal streaks on the windshield?

**Q3.14** In a rainstorm with a strong wind, what determines the best position in which to hold an umbrella?

**Q3.15** You are on the west bank of a river that is flowing north with a speed of 1.2 m/s. Your swimming speed relative to the water is 1.5 m/s, and the river is 60 m wide. What is your path relative to the earth that allows you to cross the river in the shortest time? Explain your reasoning.

**Q3.16** A stone is thrown into the air at an angle above the horizontal and feels negligible air resistance. Which graph in **Fig. Q3.16** best depicts the stone's *speed* $v$ as a function of time $t$ while it is in the air?

Figure **Q3.16**




### EXERCISES

**Section 3.1 Position and Velocity Vectors**

**3.1** • A squirrel has $x$- and $y$-coordinates (1.5 m, 2.9 m) at time $t_1 = 0$ and coordinates (5.9 m, −0.6 m) at time $t_2 = 3.0$ s. For this time interval, find (a) the components of the average velocity, and (b) the magnitude and direction of the average velocity.

**3.2** • A rhinoceros is at the origin of coordinates at time $t_1 = 0$. For the time interval from $t_1 = 0$ to $t_2 = 12.5$ s, the rhino's average velocity has $x$-component −4.0 m/s and $y$-component 4.3 m/s. At time $t_2 = 12.5$ s, (a) what are the $x$- and $y$-coordinates of the rhino? (b) How far is the rhino from the origin?

**3.3** •• **CALC** A web page designer creates an animation in which a dot on a computer screen has position

$$\vec{r} = [4.4\text{ cm} + (2.8\text{ cm/s}^2)t^2]\hat{\imath} + (5.5\text{ cm/s})t\hat{\jmath}.$$

(a) Find the magnitude and direction of the dot's average velocity between $t = 0$ and $t = 2.0$ s. (b) Find the magnitude and direction of the instantaneous velocity at $t = 0$, $t = 1.0$ s, and $t = 2.0$ s. (c) Sketch the dot's trajectory from $t = 0$ to $t = 2.0$ s, and show the velocities calculated in part (b).

**3.4** • **CALC** The position of a squirrel running in a park is given by $\vec{r} = [(0.280\text{ m/s})t + (0.0360\text{ m/s}^2)t^2]\hat{\imath} + (0.0190\text{ m/s}^3)t^3\hat{\jmath}$. (a) What are $v_x(t)$ and $v_y(t)$, the $x$- and $y$-components of the velocity of the squirrel, as functions of time? (b) At $t = 5.74$ s, how far is the squirrel from its initial position? (c) At $t = 5.74$ s, what are the magnitude and direction of the squirrel's velocity?

## Section 3.2 The Acceleration Vector

**3.5** • A jet plane is flying at a constant altitude. At time $t_1 = 0$, it has components of velocity $v_x = 88$ m/s, $v_y = 115$ m/s. At time $t_2 = 30.0$ s, the components are $v_x = -175$ m/s, $v_y = 35$ m/s. (a) Sketch the velocity vectors at $t_1$ and $t_2$. How do these two vectors differ? For this time interval calculate (b) the components of the average acceleration, and (c) the magnitude and direction of the average acceleration.

**3.6** •• A dog running in an open field has components of velocity $v_x = 3.2$ m/s and $v_y = -1.8$ m/s at $t_1 = 11.1$ s. For the time interval from $t_1 = 11.1$ s to $t_2 = 24.9$ s, the average acceleration of the dog has magnitude 0.54 m/s$^2$ and direction 33.5° measured from the $+x$-axis toward the $+y$-axis. At $t_2 = 24.9$ s, (a) what are the $x$- and $y$-components of the dog's velocity? (b) What are the magnitude and direction of the dog's velocity? (c) Sketch the velocity vectors at $t_1$ and $t_2$. How do these two vectors differ?

**3.7** •• **CALC** The coordinates of a bird flying in the $xy$-plane are given by $x(t) = \alpha t$ and $y(t) = 3.0\text{ m} - \beta t^2$, where $\alpha = 2.4$ m/s and $\beta = 1.2$ m/s$^2$. (a) Sketch the path of the bird between $t = 0$ and $t = 2.0$ s. (b) Calculate the velocity and acceleration vectors of the bird as functions of time. (c) Calculate the magnitude and direction of the bird's velocity and acceleration at $t = 2.0$ s. (d) Sketch the velocity and acceleration vectors at $t = 2.0$ s. At this instant, is the bird's speed increasing, decreasing, or not changing? Is the bird turning? If so, in what direction?

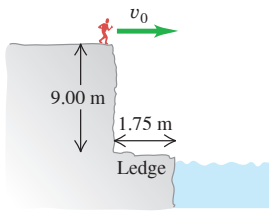
**3.8** • **CALC** A remote-controlled car is moving in a vacant parking lot. The velocity of the car as a function of time is given by $\vec{v} = [5.00\text{ m/s} - (0.0180\text{ m/s}^3)t^2]\hat{\imath} + [2.00\text{ m/s} + (0.550\text{ m/s}^2)t]\hat{\jmath}$. (a) What are $a_x(t)$ and $a_y(t)$, the $x$- and $y$-components of the car's velocity as functions of time? (b) What are the magnitude and direction of the car's velocity at $t = 6.87$ s? (c) What are the magnitude and direction of the car's acceleration at $t = 6.87$ s?

## SECTION 3.3 Projectile Motion

**3.9** • A physics book slides off a horizontal tabletop with a speed of 1.40 m/s. It strikes the floor in 0.320 s. Ignore air resistance. Find (a) the height of the tabletop above the floor; (b) the horizontal distance from the edge of the table to the point where the book strikes the floor; (c) the horizontal and vertical components of the book's velocity, and the magnitude and direction of its velocity, just before the book reaches the floor. (d) Draw $x$-$t$, $y$-$t$, $v_x$-$t$, and $v_y$-$t$ graphs for the motion.

**3.10** •• A daring 510 N swimmer dives off a cliff with a running horizontal leap, as shown in **Fig. E3.10**. What must her minimum speed be just as she leaves the top of the cliff so that she will miss the ledge at the bottom, which is 1.75 m wide and 9.00 m below the top of the cliff?

Figure **E3.10**




**3.11** • Crickets Chirpy and Milada jump from the top of a vertical cliff. Chirpy drops downward and reaches the ground in 3.10 s, while Milada jumps horizontally with an initial speed of 94.0 cm/s. How far from the base of the cliff will Milada hit the ground? Ignore air resistance.

**3.12** • You throw a basketball with an initial upward velocity component of 8.0 m/s and a horizontal velocity component of 12.0 m/s. Ignore air resistance. (a) How much time is required for the basketball to reach the highest point of the trajectory? (b) How high is this point? (c) How much time (after it is thrown) is required for the basketball to return to its original level? How does this compare with the time calculated in part (a)? (d) How far has the basketball traveled horizontally during this time? (e) Draw $x$-$t$, $y$-$t$, $v_x$-$t$, and $v_y$-$t$ graphs for the motion.

**3.13** •• **Leaping the River I.** During a storm, a car traveling on a level horizontal road comes upon a bridge that has washed out. The driver must get to the other side, so he decides to try leaping the river with his car. The side of the road the car is on is 21.5 m above the river, while the opposite side is only 2.0 m above the river. The river itself is a raging torrent 57.0 m wide. (a) How fast should the car be traveling at the time it leaves the road in order just to clear the river and land safely on the opposite side? (b) What is the speed of the car just before it lands on the other side?

**3.14** • **BIO** **The Champion Jumper of the Insect World.** The froghopper, *Philaenus spumarius,* holds the world record for insect jumps. When leaping at an angle of 58.0° above the horizontal, some of the tiny critters have reached a maximum height of 58.7 cm above the level ground. (See *Nature,* Vol. 424, July 31, 2003, p. 509.) Neglect air resistance in answering the following. (a) What was the takeoff speed for such a leap? (b) What horizontal distance did the froghopper cover for this world-record leap?

**3.15** •• Inside a starship at rest on the earth, a ball rolls off the top of a horizontal table and lands a distance $D$ from the foot of the table. This starship now lands on the unexplored Planet X. The commander, Captain Curious, rolls the same ball off the same table with the same initial speed as on earth and finds that it lands a distance $2.72D$ from the foot of the table. What is the acceleration due to gravity on Planet X?

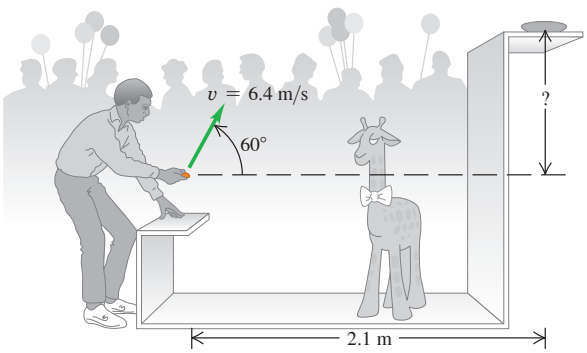
**3.16** • On level ground a shell is fired with an initial velocity of 71.0 m/s at 55.9° above the horizontal and feels no appreciable air resistance. (a) Find the horizontal and vertical components of the shell's initial velocity. (b) How long does it take the shell to reach its highest point? (c) Find its maximum height above the ground. (d) How far from its firing point does the shell land? (e) At its highest point, find the horizontal and vertical components of its acceleration and velocity.

**3.17** • A major leaguer hits a baseball so that it leaves the bat at a speed of 33.0 m/s and at an angle of 38.5° above the horizontal. Ignore air resistance. (a) At what *two* times is the baseball at a height of 10.7 m above the point at which it left the bat? (b) Calculate the horizontal and vertical components of the baseball's velocity at each of the two times calculated in part (a). (c) What are the magnitude and direction of the baseball's velocity when it returns to the level at which it left the bat?

**3.18** • A shot putter releases the shot some distance above the level ground with a velocity of 12.0 m/s, 51.0° above the horizontal. The shot hits the ground 2.08 s later. Ignore air resistance. (a) What are the components of the shot's acceleration while in flight? (b) What are the components of the shot's velocity at the beginning and at the end of its trajectory? (c) How far did she throw the shot horizontally? (d) Why does the expression for $R$ in Example 3.8 *not* give the correct answer for part (c)? (e) How high was the shot above the ground when she released it? (f) Draw $x$-$t$, $y$-$t$, $v_x$-$t$, and $v_y$-$t$ graphs for the motion.

**3.19** •• **Win the Prize.** At a funfair stall, you can win a plush giraffe toy if you toss a coin into a small dish. The dish is on a shelf above the point where the coin leaves your hand and is a horizontal distance of 2.1 m from this point (**Fig. E3.19**). If you toss the coin with a velocity of 6.4 m/s at an angle of 60° above the horizontal, the coin will land in the dish. Ignore air resistance. (a) What is the height of the shelf above the point where the coin leaves your hand? (b) What is the vertical component of the velocity of the coin just before it lands in the dish?

Figure E3.19




**3.20** • Firefighters use a high-pressure hose to shoot a stream of water at a burning building. The water has a speed of 25.0 m/s as it leaves the end of the hose and then exhibits projectile motion. The firefighters adjust the angle of elevation $\alpha$ of the hose until the water takes 3.00 s to reach a building 45.0 m away. Ignore air resistance; assume that the end of the hose is at ground level. (a) Find $\alpha$. (b) Find the speed and acceleration of the water at the highest point in its trajectory. (c) How high above the ground does the water strike the building, and how fast is it moving just before it hits the building?

**3.21** •• A man stands on the roof of a 15.0-m-tall building and throws a rock with a speed of 30.0 m/s at an angle of 33.0° above the horizontal. Ignore air resistance. Calculate (a) the maximum height above the roof that the rock reaches; (b) the speed of the rock just before it strikes the ground; and (c) the horizontal range from the base of the building to the point where the rock strikes the ground. (d) Draw $x$-$t$, $y$-$t$, $v_x$-$t$, and $v_y$-$t$ graphs for the motion.

**3.22** •• At $t = 0$ a rock is projected from ground level with a speed of 15.0 m/s and at an angle of 53.0° above the horizontal. Neglect air resistance. At what two times $t$ is the rock 5.00 m above the ground? At each of these two times, what are the horizontal and vertical components of the velocity of the rock? Let $v_{0x}$ and $v_{0y}$ be in the positive $x$- and $y$-directions, respectively.

**3.23** • Estimate the maximum horizontal distance that you can throw a basketball. (a) Based on your estimate, what is the speed of the basketball as it leaves your hand? (b) If you could throw the basketball straight up at the same speed as in part (a), how high would it go? (c) The value of $g$ on Mars is 3.7 m/s$^2$. What horizontal distance can you throw a basketball on Mars if you throw it with the same initial speed as in part (a)?

## Section 3.4 Motion in a Circle

**3.24** • Merry-go-rounds are a common ride in park playgrounds. The ride is a horizontal disk that rotates about a vertical axis at their center. A typical size is a diameter of 3.6 m. A rider sits at the outer edge of the disk and holds onto a metal bar while someone pushes on the ride to make it rotate. Estimate a typical time for one rotation. (a) For your estimated time, what is the speed of the rider, in m/s? (b) What is the rider's radial acceleration, in m/s$^2$? (c) What is the rider's radial acceleration if the time for one rotation is halved?

**3.25** • The earth has a radius of 6380 km and turns around once on its axis in 24 h. (a) What is the radial acceleration of an object at the earth's equator? Give your answer in m/s$^2$ and as a fraction of $g$. (b) If $a_{rad}$ at the equator is greater than $g$, objects will fly off the earth's surface and into space. (We'll see the reason for this in Chapter 5.) What would the period of the earth's rotation have to be for this to occur?

**3.26** •• BIO **Dizziness.** Our balance is maintained, at least in part, by the endolymph fluid in the inner ear. Spinning displaces this fluid, causing dizziness. Suppose that a skater is spinning very fast at 2.8 revolutions per second about a vertical axis through the center of his head. Take the inner ear to be approximately 7.0 cm from the axis of spin. (The distance varies from person to person.) What is the radial acceleration (in m/s$^2$ and in $g$'s) of the endolymph fluid?

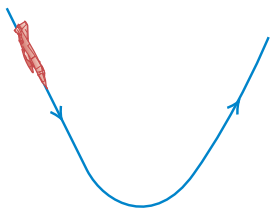
**3.27** • A small ball is attached to the lower end of a 0.800-m-long string, and the other end of the string is tied to a horizontal rod. The string makes a constant angle of 37.0° with the vertical as the ball moves at a constant speed in a horizontal circle. If it takes the ball 0.600 s to complete one revolution, what is the magnitude of the radial acceleration of the ball?

**3.28** •• A model of a helicopter rotor has four blades, each 3.00 m long from the central shaft to the blade tip. The model is rotated in a wind tunnel at 470 rev/min. (a) What is the linear speed of the blade tip, in m/s? (b) What is the radial acceleration of the blade tip expressed as a multiple of $g$?

**3.29** • BIO **Pilot Blackout in a Power Dive.** A jet plane comes in for a downward dive as shown in **Fig. E3.29.** The bottom part of the path is a quarter circle with a radius of curvature of 280 m. According to medical tests, pilots will lose consciousness when they pull out of a dive at an upward acceleration greater than 5.5$g$. At what minimum speed (in m/s and in km/h) will the pilot black out during this dive?

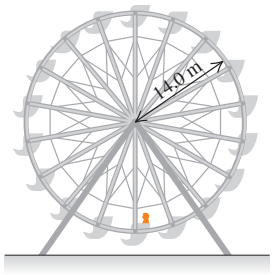
**3.30** • An object moves in a horizontal circle at constant speed $v$ (in units of m/s). It takes the object $T$ seconds to complete one revolution. Derive an expression that gives the radial acceleration of the ball in terms of $v$ and $T$, but not $r$. (a) If the speed doubles, by what factor must the period $T$ change if $a_{rad}$ is to remain unchanged? (b) If the radius doubles, by what factor must the period change to keep $a_{rad}$ the same?

Figure E3.29



**3.31** • A Ferris wheel with radius 14.0 m is turning about a horizontal axis through its center (**Fig. E3.31**). The linear speed of a passenger on the rim is constant and equal to 7.13 m/s. What are the magnitude and direction of the passenger's acceleration as she passes through (a) the lowest point in her circular motion and (b) the highest point in her circular motion? (c) How much time does it take the Ferris wheel to make one revolution?

Figure E3.31




**3.32** • A roller coaster car moves in a vertical circle of radius $R$. At the top of the circle the car has speed $v_1$, and at the bottom of the circle it has speed $v_2$, where $v_2 > v_1$. (a) When the car is at the top of its circular path, what is the direction of its radial acceleration, $a_{\text{rad, top}}$? (b) When the car is at the bottom of its circular path, what is the direction of its radial acceleration, $a_{\text{rad, bottom}}$? (c) In terms of $v_1$ and $v_2$, what is the ratio $a_{\text{rad, bottom}}/a_{\text{rad, top}}$?

**3.33** •• BIO **Hypergravity.** TsF-18 at the Yuri Gagarin Cosmonaut Training Centre, Star City, Russia, is the world's largest centrifuge. It is used to test the effects of very large accelerations ("hypergravity") on test pilots, cosmonauts, and astronauts. In this device, an arm 18.00 m long rotates about one end in a horizontal plane, and an astronaut is strapped in a seat at the other end, with their back parallel to the floor, is aligned along the centrifuge's arm. The device can reach up to $30g$, but a typical acceleration to which humans are subjected in it is between $4g$ and $9g$. (a) How fast must the astronaut's head be moving to experience a $9g$ acceleration (suppose that it is at the outermost end of the centrifuge's arm)? (b) What is the *difference* between the acceleration of their head and feet (suppose that the horizontal distance between the top of the head and the feet is 1.40 m; remember that the astronaut is seated during the process!)? (c) How fast in rpm (rev/min) is the arm turning to produce a $9g$ acceleration?

**3.34** • The radius of the earth's orbit around the sun (assumed to be circular) is $1.50 \times 10^8$ km, and the earth travels around this orbit in 365 days. (a) What is the magnitude of the orbital velocity of the earth, in m/s? (b) What is the magnitude of the radial acceleration of the earth toward the sun, in m/s$^2$? (c) Repeat parts (a) and (b) for the motion of the planet Mercury (orbit radius $= 5.79 \times 10^7$ km, orbital period $= 88.0$ days).

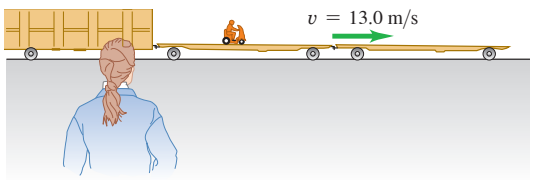
## Section 3.5 Relative Velocity

**3.35** • A travelator in an airport terminal moves at 1.4 m/s and is 32.0 m long. If a woman steps on at one end and walks at 1.7 m/s relative to the travelator how much time does it take her to reach the opposite end if she walks (a) in the same direction the travelator is moving? (b) In the opposite direction?

**3.36** • A flat wagon is traveling to the right at a speed of 13.0 m/s relative to an observer standing on the ground. Someone is riding a motor scooter on the flat wagon (**Fig. E3.36**). What is the velocity (magnitude and direction) of the scooter relative to the wagon if the scooter's velocity relative to the observer on the ground is (a) 18.0 m/s to the right? (b) 3.0 m/s to the left? (c) zero?

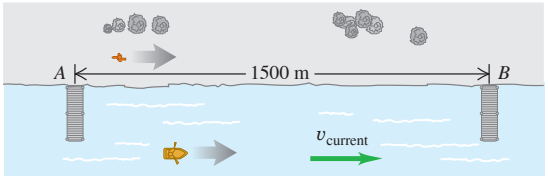
**3.37** •• A canoe has a velocity of 0.41 m/s southeast relative to the earth. The canoe is on a river that is flowing 0.60 m/s east relative to the earth. Find the velocity (magnitude and direction) of the canoe relative to the river.

Figure E3.36




**3.38** • Two piers, $A$ and $B$, are located on a river; $B$ is 1500 m downstream from $A$ (**Fig. E3.38**). Two friends must make round trips from pier $A$ to pier $B$ and return. One rows a boat at a constant speed of 4.00 km/h relative to the water; the other walks on the shore at a constant speed of 4.00 km/h. The velocity of the river is 2.80 km/h in the direction from $A$ to $B$. How much time does it take each person to make the round trip?

Figure E3.38




**3.39** •• BIO **Bird Migration.** Canada geese migrate essentially along a north–south direction for well over a thousand kilometers in some cases, traveling at speeds up to about 100 km/h. If one goose is flying at 100 km/h relative to the air but a 50 km/h wind is blowing from west to east, (a) at what angle relative to the north–south direction should this bird head to travel directly southward relative to the ground? (b) How long will it take the goose to cover a ground distance of 550 km from north to south? (*Note:* Even on cloudy nights, many birds can navigate by using the earth's magnetic field to fix the north–south direction.)

**3.40** •• The nose of an ultralight plane is pointed due south, and its airspeed indicator shows 35 m/s. The plane is in a 10 m/s wind blowing toward the southwest relative to the earth. (a) In a vector-addition diagram, show the relationship of $\vec{v}_{P/E}$ (the velocity of the plane relative to the earth) to the two given vectors. (b) Let $x$ be east and $y$ be north, and find the components of $\vec{v}_{P/E}$. (c) Find the magnitude and direction of $\vec{v}_{P/E}$.

**3.41** • **Crossing the River I.** A river flows due south with a speed of 2.0 m/s. You steer a motorboat across the river; your velocity relative to the water is 4.2 m/s due east. The river is 500 m wide. (a) What is your velocity (magnitude and direction) relative to the earth? (b) How much time is required to cross the river? (c) How far south of your starting point will you reach the opposite bank?

**3.42** • **Crossing the River II.** (a) In which direction should the motorboat in Exercise 3.41 head to reach a point on the opposite bank directly east from your starting point? (The boat's speed relative to the water remains 4.2 m/s.) (b) What is the velocity of the boat relative to the earth? (c) How much time is required to cross the river?

**3.43** •• An airplane pilot wishes to fly due west. A wind of 80.0 km/h is blowing toward the south. (a) If the airspeed of the plane (its speed in still air) is 320.0 km/h, in which direction should the pilot head? (b) What is the speed of the plane over the ground? Draw a vector diagram.

## PROBLEMS

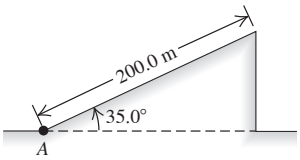
**3.44** •• CP CALC A dog in an open field is at rest under a tree at time $t = 0$ and then runs with acceleration $\vec{a}(t) = (0.400\ \text{m/s}^2)\hat{\imath} - (0.180\ \text{m/s}^3)t\hat{\jmath}$. How far is the dog from the tree 8.00 s after it starts to run?

**3.45** •• CALC If $\vec{r} = bt^2\hat{\imath} + ct^3\hat{\jmath}$, where $b$ and $c$ are positive constants, when does the velocity vector make an angle of 45.0° with the $x$- and $y$-axes?

**3.46** ••• **CALC** A faulty model rocket moves in the $xy$-plane (the positive $y$-direction is vertically upward). The rocket's acceleration has components $a_x(t) = \alpha t^2$ and $a_y(t) = \beta - \gamma t$, where $\alpha = 2.50\ \text{m/s}^4$, $\beta = 9.00\ \text{m/s}^2$, and $\gamma = 1.40\ \text{m/s}^3$. At $t = 0$ the rocket is at the origin and has velocity $\vec{v}_0 = v_{0x}\hat{\imath} + v_{0y}\hat{\jmath}$ with $v_{0x} = 1.00\ \text{m/s}$ and $v_{0y} = 7.00\ \text{m/s}$. (a) Calculate the velocity and position vectors as functions of time. (b) What is the maximum height reached by the rocket? (c) What is the horizontal displacement of the rocket when it returns to $y = 0$?

**3.47** ••• **CP** A test rocket starting from rest at point $A$ is launched by accelerating it along a 200.0-m incline at $1.48\ \text{m/s}^2$ (**Fig. P3.47**). The incline rises at 35.0° above the horizontal, and at the instant the rocket leaves it, the engines turn off and the rocket is subject to gravity only (ignore air resistance). Find (a) the maximum height above the ground that the rocket reaches, and (b) the rocket's greatest horizontal range beyond point $A$.

Figure **P3.47**




**3.48** •• **CALC** The position of a dragonfly that is flying parallel to the ground is given as a function of time by $\vec{r} = [2.90\ \text{m} + (0.0900\ \text{m/s}^2)t^2]\hat{\imath} - (0.0150\ \text{m/s}^3)t^3\hat{\jmath}$. (a) At what value of $t$ does the velocity vector of the dragonfly make an angle of 32.0° clockwise from the $+x$-axis? (b) At the time calculated in part (a), what are the magnitude and direction of the dragonfly's acceleration vector?

**3.49** •• In fighting forest fires, airplanes work in support of ground crews by dropping water on the fires. For practice, a pilot drops a canister of red dye, hoping to hit a target on the ground below. If the plane is flying in a horizontal path 90.0 m above the ground and has a speed of 62.0 m/s at what horizontal distance from the target should the pilot release the canister? Ignore air resistance.

**3.50** •• **CALC** A bird flies in the $xy$-plane with a velocity vector given by $\vec{v} = (\alpha - \beta t^2)\hat{\imath} + \gamma t\hat{\jmath}$, with $\alpha = 2.4\ \text{m/s}$, $\beta = 1.6\ \text{m/s}^3$, and $\gamma = 4.0\ \text{m/s}^2$. The positive $y$-direction is vertically upward. At $t = 0$ the bird is at the origin. (a) Calculate the position and acceleration vectors of the bird as functions of time. (b) What is the bird's altitude ($y$-coordinate) as it flies over $x = 0$ for the first time after $t = 0$?

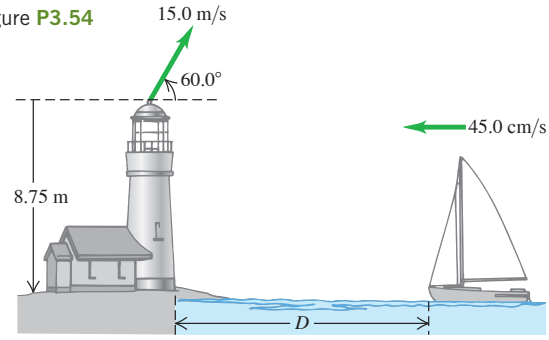
**3.51** ••• A movie stuntwoman drops from a helicopter that is 30.0 m above the ground and moving with a constant velocity whose components are 10.0 m/s upward and 15.0 m/s horizontal and toward the south. Ignore air resistance. (a) Where on the ground (relative to the position of the helicopter when she drops) should the stuntwoman have placed foam mats to break her fall? (b) Draw $x$-$t$, $y$-$t$, $v_x$-$t$, and $v_y$-$t$ graphs of her motion.

**3.52** •• A cannon, located 60.0 m from the base of a vertical 25.0-m-tall cliff, shoots a 15 kg shell at 43.0° above the horizontal toward the cliff. (a) What must the minimum muzzle velocity be for the shell to clear the top of the cliff? (b) The ground at the top of the cliff is level, with a constant elevation of 25.0 m above the cannon. Under the conditions of part (a), how far does the shell land past the edge of the cliff?

**3.53** • **CP** **CALC** A toy rocket is launched with an initial velocity of 15.0 m/s in the horizontal direction from the roof of a 39.0-m-tall building. The rocket's engine produces a horizontal acceleration of $(1.60\ \text{m/s}^3)t$, in the same direction as the initial velocity, but in the vertical direction the acceleration is $g$, downward. Ignore air resistance. What horizontal distance does the rocket travel before reaching the ground?

**3.54** ••• An important piece of landing equipment must be thrown to a ship, which is moving at 45.0 cm/s, before the ship can dock. This equipment is thrown at 15.0 m/s at 60.0° above the horizontal from the top of a tower at the edge of the water, 8.75 m above the ship's deck (**Fig. P3.54**). For this equipment to land at the front of the ship, at what distance $D$ from the dock should the ship be when the equipment is thrown? Ignore air resistance.

Figure **P3.54**

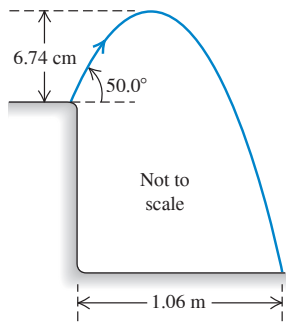



**3.55** •• A cricket ball thrown at an angle of 60.0° above the horizontal strikes a building 18.0 m away at a point 8.00 m above the point from which it is thrown. Ignore air resistance. (a) Find the magnitude of the ball's initial velocity (the velocity with which the ball is thrown). (b) Find the magnitude and direction of the velocity of the ball just before it strikes the building.

**3.56** •• **An Errand of Mercy.** An airplane is dropping bales of hay to cattle stranded in a blizzard on the Great Plains. The pilot releases the bales at 180 m above the level ground when the plane is flying at 80 m/s in a direction 60° above the horizontal. How far in front of the cattle should the pilot release the hay so that the bales land at the point where the cattle are stranded?

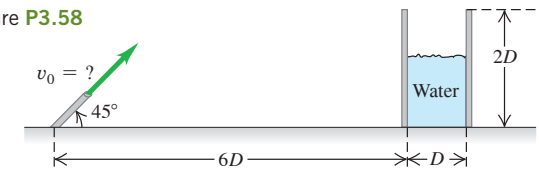
**3.57** •• A grasshopper leaps into the air from the edge of a vertical cliff, as shown in **Fig. P3.57.** Find (a) the initial speed of the grasshopper and (b) the height of the cliff.

Figure **P3.57**




**3.58** ••• A water hose is used to fill a large cylindrical storage tank of diameter $D$ and height $2D$. The hose shoots the water at 45° above the horizontal from the same level as the base of the tank and is a distance $6D$ away (**Fig. P3.58**). For what *range* of launch speeds ($v_0$) will the water enter the tank? Ignore air resistance, and express your answer in terms of $D$ and $g$.

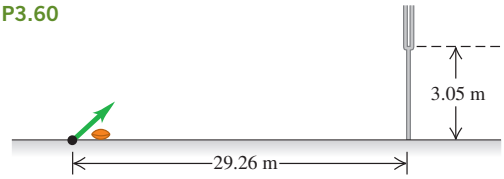
Figure **P3.58**



**3.59** •• An object is projected with initial speed $v_0$ from the edge of the roof of a building that has height $H$. The initial velocity of the object makes an angle $\alpha_0$ with the horizontal. Neglect air resistance. (a) If $\alpha_0$ is 90°, so that the object is thrown straight up (but misses the roof on the way down), what is the speed $v$ of the object just before it strikes the ground? (b) If $\alpha_0 = -90°$, so that the object is thrown straight down, what is its speed just before it strikes the ground? (c) Derive an expression for the speed $v$ of the object just before it strikes the ground for general $\alpha_0$. (d) The final speed $v$ equals $v_1$ when $\alpha_0$ equals $\alpha_1$. If $\alpha_0$ is increased, does $v$ increase, decrease, or stay the same?

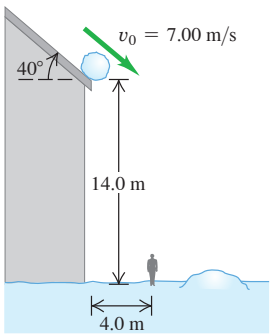
**3.60** •• **Kicking an Extra Point.** In Canadian football, after a touchdown the team has the opportunity to earn one more point by kicking the ball over the bar between the goal posts. Football regulations are stated in British units (see Appendix C): the bar is 10 feet (3.05 m) above the ground, and the ball is kicked from ground level, 32 yards (29.26 m) horizontally from the bar (**Fig. P3.60**). (a) There is a minimum angle above the ground such that if the ball is launched below this angle, it can never clear the bar, no matter how fast it is kicked. What is this angle? (b) If the ball is kicked at 45.0° above the horizontal, what must its initial speed be if it is just to clear the bar? Express your answer in m/s and in km/h.

Figure **P3.60**




**3.61** ••• **Look Out!** A snowball rolls off a barn roof that slopes downward at an angle of 40° (**Fig. P3.61**). The edge of the roof is 14.0 m above the ground, and the snowball has a speed of 7.00 m/s as it rolls off the roof. Ignore air resistance. (a) How far from the edge of the barn does the snowball strike the ground if it doesn't strike anything else while falling? (b) Draw $x$-$t$, $y$-$t$, $v_x$-$t$, and $v_y$-$t$ graphs for the motion in part (a). (c) A man 1.9 m tall is standing 4.0 m from the edge of the barn. Will the snowball hit him?

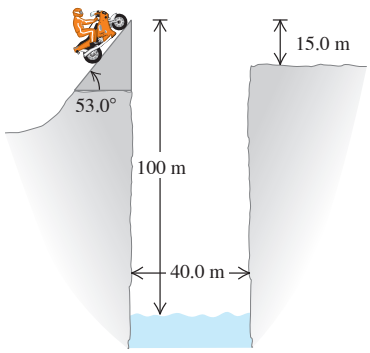
Figure **P3.61**




**3.62** • A 2.7 kg ball is thrown upward with an initial speed of 20.0 m/s from the edge of a 45.0-m-high cliff. At the instant the ball is thrown, a woman starts running away from the base of the cliff with a constant speed of 6.00 m/s. The woman runs in a straight line on level ground. Ignore air resistance on the ball. (a) At what angle above the horizontal should the ball be thrown so that the runner will catch it just before it hits the ground, and how far does she run before she catches the ball? (b) Carefully sketch the ball's trajectory as viewed by (i) a person at rest on the ground and (ii) the runner.

**3.63** •• **Leaping the River II.** A physics professor did daredevil stunts in his spare time. His last stunt was an attempt to jump across a river on a motorcycle (**Fig. P3.63**). The takeoff ramp was inclined at 53.0°, the river was 40.0 m wide, and the far bank was 15.0 m lower than the top of the ramp. The river itself was 100 m below the ramp. Ignore air resistance. (a) What should his speed have been at the top of the ramp to have just made it to the edge of the far bank? (b) If his speed was only half the value found in part (a), where did he land?

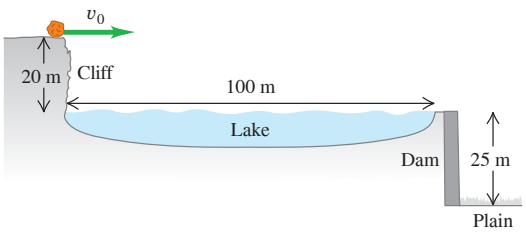
Figure **P3.63**




**3.64** •• **Tossing Your Lunch.** Henrietta is jogging along the pavement at 3.0 m/s on the way to her physics class. Bruce realizes that she forgot her bag of bagels, so he runs to the window, which is 43.8 m above the street level and directly above the pavement, to throw the bag to her. He throws it horizontally 9.00 s after she has passed below the window, and she catches it on the run. Ignore air resistance. (a) With what initial speed must Bruce throw the bagels so that Henrietta can catch the bag just before it hits the ground? (b) Where is Henrietta when she catches the bagels?

**3.65** • A 76.0 kg rock is rolling horizontally at the top of a vertical cliff that is 20 m above the surface of a lake (**Fig. P3.65**). The top of the vertical face of a dam is located 100 m from the foot of the cliff, with the top of the dam level with the surface of the water in the lake. A level plain is 25 m below the top of the dam. (a) What must be the minimum speed of the rock just as it leaves the cliff so that it will reach the plain without striking the dam? (b) How far from the foot of the dam does the rock hit the plain?

Figure **P3.65**




**3.66** •• A firefighting crew uses a water cannon that shoots water at 25.0 m/s at a fixed angle of 53.0° above the horizontal. The firefighters want to direct the water at a blaze that is 10.0 m above ground level. How far from the building should they position their cannon? There are *two* possibilities; can you get them both? (*Hint:* Start with a sketch showing the trajectory of the water.)

**3.67** •• On a level football pitch a football is projected from ground level. It has speed 8.0 m/s when it is at its maximum height. It travels a horizontal distance of 50.0 m. Neglect air resistance. How long is the ball in the air?

**3.68** ••• You are standing on a loading dock at the top of a flat ramp that is at a constant angle $\alpha_0$ below the horizontal. You slide a small box horizontally off the loading dock with speed $v_0$ in a direction so that it lands on the ramp. How far vertically downward does the box travel before it strikes the ramp?

**3.69** ••• In the middle of the night you are standing a horizontal distance of 14.0 m from the high fence that surrounds the estate of your rich uncle. The top of the fence is 5.00 m above the ground. You have taped an important message to a rock that you want to throw over the fence. The ground is level, and the width of the fence is small enough to be ignored. You throw the rock from a height of 1.60 m above the ground and at an angle of 56.0° above the horizontal. (a) What minimum initial speed must the rock have as it leaves your hand to clear the top of the fence? (b) For the initial velocity calculated in part (a), what horizontal distance beyond the fence will the rock land on the ground?

**3.70** ••• A small object is projected from level ground with an initial velocity of magnitude 16.0 m/s and directed at an angle of 60.0° above the horizontal. (a) What is the horizontal displacement of the object when it is at its maximum height? How does your result compare to the horizontal range $R$ of the object? (b) What is the vertical displacement of the object when its horizontal displacement is 80.0% of its horizontal range $R$? How does your result compare to the maximum height $h_{max}$ reached by the object? (c) For when the object has horizontal displacement $x - x_0 = \alpha R$, where $\alpha$ is a positive constant, derive an expression (in terms of $\alpha$) for $(y - y_0)/h_{max}$. Your result should not depend on the initial velocity or the angle of projection. Show that your expression gives the correct result when $\alpha = 0.80$, as is the case in part (b). Also show that your expression gives the correct result for $\alpha = 0$, $\alpha = 0.50$, and $\alpha = 1.0$.

**3.71** •• An airplane pilot sets a compass course due west and maintains an airspeed of 214 km/h. After flying for 0.470 h, she finds herself over a town 123 km west and 12 km south of her starting point. (a) Find the wind velocity (magnitude and direction). (b) If the wind velocity is 40 km/h due south, in what direction should the pilot set her course to travel due west? Use the same airspeed of 214 km/h.

**3.72** •• **Raindrops.** When a train's velocity is 12.0 m/s eastward, raindrops that are falling vertically with respect to the earth make traces that are inclined 30.0° to the vertical on the windows of the train. (a) What is the horizontal component of a drop's velocity with respect to the earth? With respect to the train? (b) What is the magnitude of the velocity of the raindrop with respect to the earth? With respect to the train?

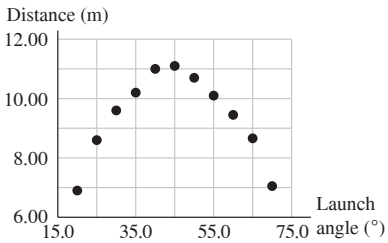
**3.73** ••• In a World Cup football match, Juan is running due north toward the goal with a speed of 7.30 m/s relative to the ground. A teammate passes the ball to him. The ball has a speed of 13.3 m/s and is moving in a direction 30.6° east of north, relative to the ground. What are the magnitude and direction of the ball's velocity relative to Juan?

**3.74** •• A shortstop is running due east as he throws a baseball to the catcher, who is standing at home plate. The velocity of the baseball relative to the shortstop is 6.00 m/s in the direction due south, and the speed of the baseball relative to the catcher is 9.00 m/s. What is the speed of the shortstop relative to the ground when he throws the ball?

**3.75** •• Two football players, Mia and Alice, are running as Alice passes the ball to Mia. Mia is running due north with a speed of 6.30 m/s. The velocity of the ball relative to Mia is 6.80 m/s in a direction 30.0° east of south. What are the magnitude and direction of the velocity of the ball relative to the ground?

**3.76** •• **DATA** A spring-gun projects a small rock from the ground with speed $v_0$ at an angle $\theta_0$ above the ground. You have been asked to determine $v_0$. From the way the spring-gun is constructed, you know that to a good approximation $v_0$ is independent of the launch angle. You go to a level, open field, select a launch angle, and measure the horizontal distance the rock travels. You use $g = 9.80\ \text{m/s}^2$ and ignore the small height of the end of the spring-gun's barrel above the ground. Since your measurement includes some uncertainty in values measured for the launch angle and for the horizontal range, you repeat the measurement for several launch angles and obtain the results given in **Fig. 3.76**. You ignore air resistance because there is no wind and the rock is small and heavy. (a) Select a way to represent the data well as a straight line. (b) Use the slope of the best straight-line fit to your data from part (a) to calculate $v_0$. (c) When the launch angle is 36.9°, what maximum height above the ground does the rock reach?

Figure **P3.76**




**3.77** •• **DATA** You have constructed a hair-spray-powered potato gun and want to find the muzzle speed $v_0$ of the potatoes, the speed they have as they leave the end of the gun barrel. You use the same amount of hair spray each time you fire the gun, and you have confirmed by repeated firings at the same height that the muzzle speed is approximately the same for each firing. You climb on a microwave relay tower (with permission, of course) to launch the potatoes horizontally from different heights above the ground. Your friend measures the height of the gun barrel above the ground and the range $R$ of each potato. You obtain the data in the table.

| Launch height $h$ | Horizontal range $R$ |
|---|---|
| 2.00 m | 10.4 m |
| 6.00 m | 17.1 m |
| 9.00 m | 21.3 m |
| 12.00 m | 25.8 m |

Each of the values of $h$ and $R$ has some measurement error: The muzzle speed is not precisely the same each time, and the barrel isn't precisely horizontal. So you use all of the measurements to get the best estimate of $v_0$. No wind is blowing, so you decide to ignore air resistance. You use $g = 9.80\ \text{m/s}^2$ in your analysis. (a) Select a way to represent the data well as a straight line. (b) Use the slope of the best-fit line from part (a) to calculate the average value of $v_0$. (c) What would be the horizontal range of a potato that is fired from ground level at an angle of 30.0° above the horizontal? Use the value of $v_0$ that you calculated in part (b).

**3.78** ••• **DATA** You are a member of a geological team in Central Africa. Your team comes upon a wide river that is flowing east. You must determine the width of the river and the current speed (the speed of the water relative to the earth). You have a small boat with an outboard motor. By measuring the time it takes to cross a pond where the water isn't flowing, you have calibrated the throttle settings to the speed of the boat in still water. You set the throttle so that the speed of the boat relative to the river is a constant 6.00 m/s. Traveling due north across the river, you reach the opposite bank in 20.1 s. For the return trip, you change the throttle setting so that the speed of the boat relative to the water is 9.00 m/s. You travel due south from one bank to the other and cross the river in 11.2 s. (a) How wide is the river, and what is the current speed? (b) With the throttle set so that the speed of the boat relative to the water is 6.00 m/s, what is the shortest time in which you could cross the river, and where on the far bank would you land?

## CHALLENGE PROBLEMS

**3.79** ••• **CALC** A projectile thrown from a point $P$ moves in such a way that its distance from $P$ is always increasing. Find the maximum angle above the horizontal with which the projectile could have been thrown. Ignore air resistance.

**3.80** ••• Two students are canoeing on a river. While heading upstream, they accidentally drop an empty bottle overboard. They then continue paddling for 60 minutes, reaching a point 2.0 km farther upstream. At this point they realize that the bottle is missing and, driven by ecological awareness, they turn around and head downstream. They catch up with and retrieve the bottle (which has been moving along with the current) 5.0 km downstream from the turnaround point. (a) Assuming a constant paddling effort throughout, how fast is the river flowing? (b) What would the canoe speed in a still lake be for the same paddling effort?

**3.81** ••• **CP** A rocket designed to place small payloads into orbit is carried to an altitude of 12.0 km above sea level by a converted airliner. When the airliner is flying in a straight line at a constant speed of 850 km/h, the rocket is dropped. After the drop, the airliner maintains the same altitude and speed and continues to fly in a straight line. The rocket falls for a brief time, after which its rocket motor turns on. Once that motor is on, the combined effects of thrust and gravity give the rocket a constant acceleration of magnitude $3.00g$ directed at an angle of 30.0° above the horizontal. For safety, the rocket should be at least 1.00 km in front of the airliner when it climbs through the airliner's altitude. Your job is to determine the minimum time that the rocket must fall before its engine starts. Ignore air resistance. Your answer should include (i) a diagram showing the flight paths of both the rocket and the airliner, labeled at several points with vectors for their velocities and accelerations; (ii) an $x$-$t$ graph showing the motions of both the rocket and the airliner; and (iii) a $y$-$t$ graph showing the motions of both the rocket and the airliner. In the diagram and the graphs, indicate when the rocket is dropped, when the rocket motor turns on, and when the rocket climbs through the altitude of the airliner.

## MCAT-STYLE PASSAGE PROBLEMS

**BIO Ballistic Seed Dispersal.** Some plants disperse their seeds when the fruit splits and contracts, propelling the seeds through the air. The trajectory of these seeds can be determined with a high-speed camera. In an experiment on one type of plant, seeds are projected at 20 cm above ground level with initial speeds between 2.3 m/s and 4.6 m/s. The launch angle is measured from the horizontal, with +90° corresponding to an initial velocity straight up and −90° straight down.

**3.82** The experiment is designed so that the seeds move no more than 0.20 mm between photographic frames. What minimum frame rate for the high-speed camera is needed to achieve this? (a) 250 frames/s; (b) 2500 frames/s; (c) 25,000 frames/s; (d) 250,000 frames/s.

**3.83** About how long does it take a seed launched at 90° at the highest possible initial speed to reach its maximum height? Ignore air resistance. (a) 0.23 s; (b) 0.47 s; (c) 1.0 s; (d) 2.3 s.

**3.84** If a seed is launched at an angle of 0° with the maximum initial speed, how far from the plant will it land? Ignore air resistance, and assume that the ground is flat. (a) 20 cm; (b) 93 cm; (c) 2.2 m; (d) 4.6 m.

**3.85** A large number of seeds are observed, and their initial launch angles are recorded. The range of projection angles is found to be −51° to 75°, with a mean of 31°. Approximately 65% of the seeds are launched between 6° and 56°. (See W. J. Garrison et al., "Ballistic seed projection in two herbaceous species," *Amer. J. Bot.,* Sept. 2000, 87:9, 1257–64.) Which of these hypotheses is best supported by the data? Seeds are preferentially launched (a) at angles that maximize the height they travel above the plant; (b) at angles below the horizontal in order to drive the seeds into the ground with more force; (c) at angles that maximize the horizontal distance the seeds travel from the plant; (d) at angles that minimize the time the seeds spend exposed to the air.

# ANSWERS

## Chapter Opening Question ?

**(iii)** A cyclist going around a curve at constant speed has an acceleration directed toward the inside of the curve (see Section 3.2, especially Fig. 3.12a).

## Key Example VARIATION Problems

**VP3.9.1** (a) 1.53 s, 11.5 m (b) 3.06 s, 61.2 m
**VP3.9.2** (a) 20.6 m/s (b) 5.40 m
**VP3.9.3** (a) 3.51 s (b) 33.8 m (c) $v_x = 9.64$ m/s, $v_y = -22.9$ m/s
**VP3.9.4** (a) $t = 2v_0/g$, $x = 2v_0^2/g$, $y = -2v_0^2/g$ (b) $t = v_0/g$, $x = v_0^2/g$, $y = -v_0^2/2g$
**VP3.12.1** 20.0 m
**VP3.12.2** (a) 41.6 s (b) 6.04 m/s$^2$
**VP3.12.3** (a) $P$: 0.938 m/s, $Q$: 1.88 m/s (b) $P$: 8.79 m/s$^2$, $Q$: 17.6 m/s$^2$ (c) increase, increase
**VP3.12.4** (a) Venus: $3.49 \times 10^4$ m/s, Earth: $2.99 \times 10^4$ m/s, Mars: $2.41 \times 10^4$ m/s (b) Venus: $1.13 \times 10^{-2}$ m/s$^2$, Earth: $5.95 \times 10^{-3}$ m/s$^2$, Mars: $2.55 \times 10^{-3}$ m/s$^2$ (c) decrease, decrease
**VP3.12.5** (a) $T_A/T_B = 4$ (b) $a_{\text{rad},A}/a_{\text{rad},B} = 1/8$
**VP3.15.1** (a) +53.0 m/s (b) −53.0 m/s
**VP3.15.2** (a) 90.0 m/s eastward (b) 90.0 m/s westward (c) neither
**VP3.15.3** (a) 25.6 m/s, 51.3° north of east (b) 25.6 m/s, 51.3° south of west
**VP3.15.4** 130 m/s, 12.6° west of north

## Bridging Problem

(a) $R = \dfrac{2v_0^2}{g}\dfrac{\cos(\theta + \phi)\sin\phi}{\cos^2\theta}$ (b) $\phi = 45° - \dfrac{\theta}{2}$